



Service Tool Instruction

Service Tool Instruction
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2003

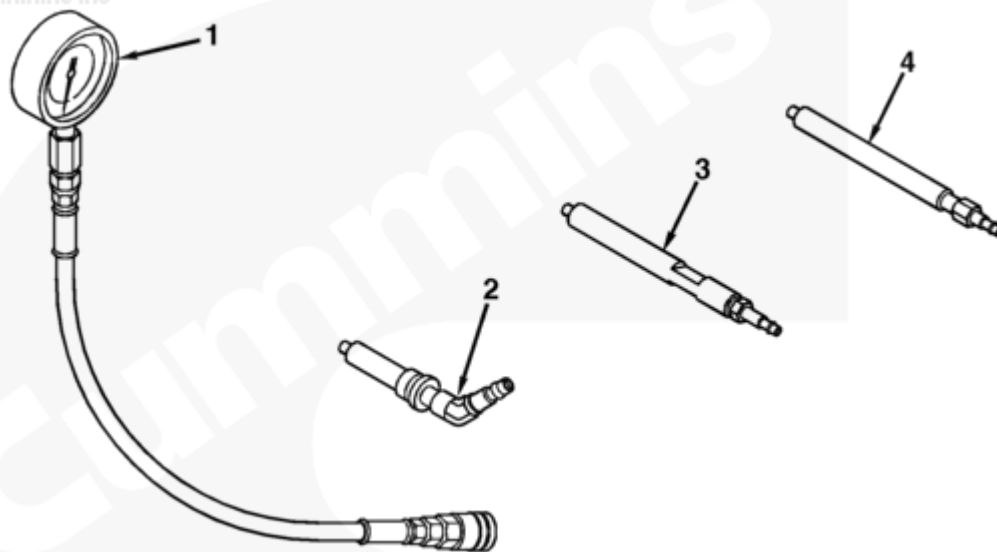
Description

Compression Gauge

Purpose

This document provides information for using compression gauge, Part Number 3164627. The gauge is designed for use on diesel engines and has a test range of 0 to 5516 kPa [0 to 800 psi] that is divided into 100 kPa [20 psi] increments. The compression gauge is reset to zero with a push button release valve. The gauge is used to measure cylinder pressure (at cranking speed) on the A1400, A1700, A2000, A2300, A2300T, B3.3, B3.9, QSB3.9-30, QSB4.5-30, B5.9, QSB5.9-30, B4.5^s, B6.7^s and C8.3 series engines.

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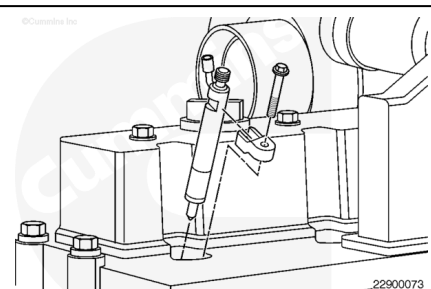
Table 1, Compression Gauge, Part Number 3164627

Item Number	Part Number	Description	Quantity
1	3164627	Compression gauge	1

Table 2, Items Used with the Compression Gauge, Purchased Separately			
Item Number	Part Number	Description	Quantity
2	3164839	B series compression tester (2-valve)	1
3	3164728	B3.3 series compression tester	1
4	3164840	C series compression tester (2-valve)	1
Not shown	3165128	A1400, A1700, A2000, A2300, and A2300T compression tester	1

Remove all injectors. Refer to the following Troubleshooting and Repair or Master Repair Manuals:

- Troubleshooting and Repair Manual B Series Engines 1991 and 1994 Certification Levels, Bulletin 3666087
(/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666087.html)
- Troubleshooting and Repair Manual B Series Engines, Bulletin 3810207 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3810207.html)
- Troubleshooting and Repair Manual B3.3 Engines, Bulletin 3666418 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666418.html)
- Troubleshooting and Repair Manual C Series Engines 1991 and 1994 Certification Levels, Bulletin 3666003
(/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666003.html)
- Troubleshooting and Repair Manual C Series Engines, Bulletin 3810261 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3810261.html)
- Master Repair Manual A Series, Bulletin 4021433
(/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021433.html).



⚠ CAUTION ⚠

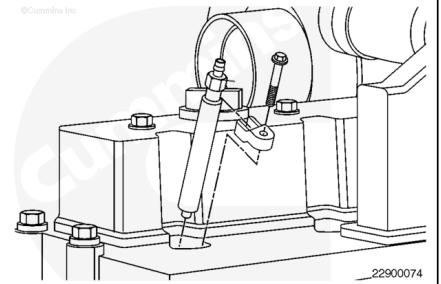
Make certain that no fuel or oil remain in the combustion chamber after the injectors have been removed. If fuel or oil remain in the cylinder during the compression test, damage

to the compression gauge can result.

Temporarily disable the fuel injection system.

Rotate the engine using the starter to remove any remaining fuel or oil from the combustion chamber. Install the appropriate compression tester into the injector bore. For the C Series engine, install the injector clamp and capscrew.

Torque Value: 20 n·m [180 in-lb]



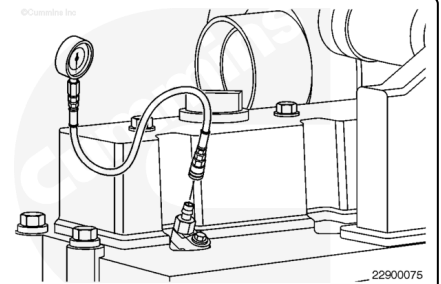
⚠ WARNING ⚠

Press the release button on the compression gauge to release the pressure before removing the gauge from the engine. Failure to do so can cause personal injury.

Connect the compression gauge to the compression tester.

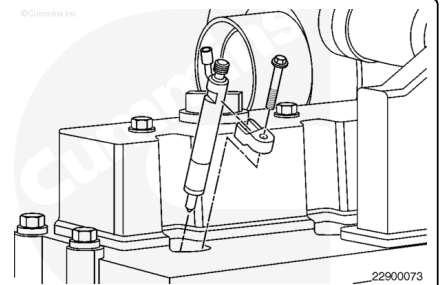
Refer to the appropriate Troubleshooting and Repair or Master Repair Manual for the compression testing procedure.

Disconnect the compression gauge from the compression tester. Remove the compression tester. For the C Series engine, remove the capscrew, injector clamp, and compression tester.



Install all injectors. Refer to the following Troubleshooting and Repair or Master Repair Manuals:

- Troubleshooting and Repair Manual B Series Engines 1991 and 1994 Certification Levels, Bulletin 3666087 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666087.html>)
- Troubleshooting and Repair Manual B Series Engines, Bulletin 3810207 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3810207.html>)
- Troubleshooting and Repair Manual B3.3 Engines, Bulletin 3666418 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666418.html>)
- Troubleshooting and Repair Manual C Series Engines 1991 and 1994 Certification Levels, Bulletin 3666003 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666003.html>)
- Troubleshooting and Repair Manual C Series Engines, Bulletin 3810261 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3810261.html>)



- Master Repair Manual A Series, Bulletin 4021433
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